

Multimodal Deep Learning Based Crop Classification Using Multispectral and Multitemporal Satellite Imagery

Krishna Karthik Gadiraju
kgadira@ncsu.edu
North Carolina State University
Raleigh, North Carolina

Zexi Chen
zchen22@ncsu.edu
North Carolina State University
Raleigh, North Carolina

Bharathkumar Ramachandra
bramach2@ncsu.edu
North Carolina State University
Raleigh, North Carolina

Ranga Raju Vatsavai
rrvatsav@ncsu.edu
North Carolina State University
Raleigh, North Carolina

ABSTRACT

The Food and Agriculture Organization (FAO) of the United Nations predicts that in order to meet the needs of the expected 3 billion population growth by 2050, food production has to increase by 60%. Therefore, monitoring and mapping crops accurately is essential for estimating food production during each crop growing season across the globe. Traditionally, multispectral remote sensing imagery has been widely used for mapping crops worldwide. However, single date imagery does not capture temporal characteristics (phenology) of growing crops, leading to imprecise crop maps and food estimates. On the other hand, purely temporal classification approaches also produce inaccurate crop maps as they do not account for spatial autocorrelations. In this paper, we present a multimodal deep learning solution that jointly exploits spatial-spectral and phenological properties to identify major crop types. Using a two stream architecture, spatial characteristics are captured via a spatial stream consisting of very high resolution images (single date, 1m, 3-spectral bands, USDA NAIP) with a CNN and the phenological characteristics via a temporal stream images (biweekly, 250m, MODIS NDVI) with an LSTM. Experimental results show that the proposed multimodal solution reduces prediction error by 60%.

CCS CONCEPTS

• **Information systems** → **Spatial-temporal systems**; • **Computing methodologies** → **Machine learning**; **Supervised learning**; **Neural networks**; • **Applied computing** → **Agriculture**.

KEYWORDS

neural networks, crop classification, remote sensing, multimodal

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1 INTRODUCTION

It is estimated that 821 million people in the world suffered from hunger in 2018. Population growth coupled with the corresponding increase in demand for supply of food is a major challenge for governments across the world. According to the United Nations Food and Agricultural Organization (FAO) [12], 60% more food needs to be produced in order to sustain the needs of the world population by 2050. Accurate and automatic identification of different types of crops would assist the governments in preparing for this sustainability challenge. Machine learning solutions for crop classification using the readily available large scale remote sensing imagery could provide a quick and cost effective solution for crop monitoring and global food estimation.

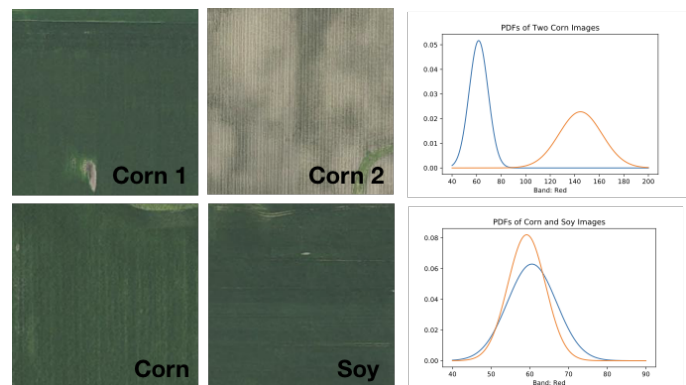


Figure 1: Inter- vs. Intra class variability. Top row shows two corn fields that are very dissimilar leading high variability and bottom row shows corn vs soybean fields that are spectrally similar. Corresponding one dimensional pdf plots illustrate this point.

Crop classification using remote sensing imagery has several challenges such as high intra-class variability and low inter-class variability (see Figure 1). For instance, high intra-class variability can be observed when images of a corn field are acquired at different periods of its growth cycle. Similarly, low inter-class variance can be observed between images belonging to different classes, such as corn and soy are collected during similar growth cycle. Methods that use single-date imagery and per-pixel classification methods such as [4, 6, 27, 31] (e.g. Support Vector Machines (SVM) and Maximum Likelihood, Random Forests) and object based classification methods such as [8, 22] are susceptible to the aforementioned challenges. While very high resolution (VHR) imagery such as NAIP are highly useful in crop classification, they are collected once a year at the most. As a result, crop classification schemes using single date imagery, will still suffer from high inter-class and low intra-class variability even if the data is collected during the best date in a crop growing season.

On the other hand high-temporal images such as MODIS [20] allow us to accurately model crop phenology, but they suffer from low spatial resolution (250 m). Many large scale mapping efforts such as [14] have widely used “Normalized Difference Vegetation Index” (NDVI) product from the MODIS satellite. NDVI is calculated using the Near Infrared (NIR) and Infrared (IR) bands of a multispectral image using the formula $NDVI = \frac{NIR - IR}{NIR + IR}$ and is one of the most often used method for vegetation monitoring. However, it is possible that the growth cycle of two different crops in terms of their NDVI features may also look very similar. In such cases, it may not be possible to discriminate such classes without good spatial and spectral information. For example, consider the plots shown in Figure 2 containing time series plots of 1000 randomly selected time series of NDVI values for the crops corn, cotton, soy, spring wheat, winter wheat and barley. Within each plot, the average value per time stamp for all of the randomly selected time series is represented in red. From the plots, we can see high intra-class similarity between corn and soy and between spring wheat and barley.

Typical crop classification projects are either local (confined to small regions with fewer crop varieties) or global. Local crop mapping efforts are highly curated in-terms of image selection (typically using medium resolution images, such as Landsat TM/ETM+) and temporal sampling during crop growing season to avoid some of the problems described above. On the other hand, global land cover mapping efforts are using mostly high temporal images with coarse spatial resolution (e.g., MODIS NDVI). Examples include: (i) GLC2000 [33] which was developed by European Commission’s Joint Research Center at 1km spatial resolution using SPOT-VEGETATION daily data, (ii) GlobCover [1] was developed by European Space Agency (ESA) and its partners using MERIS bi-monthly time series imagery at 300m spatial resolution, and (iii) in the USA, Boston University and its partners developed global MODISLC [14] classification data products using MODIS data at 500m spatial resolution. Most of these global efforts with time series images have reached an accuracy of 70 to 90%.

Our work tries to overcome these limitations by jointly exploiting the spectral and spatial characteristics from very high-resolution images and crop phenology properties captured by high-temporal

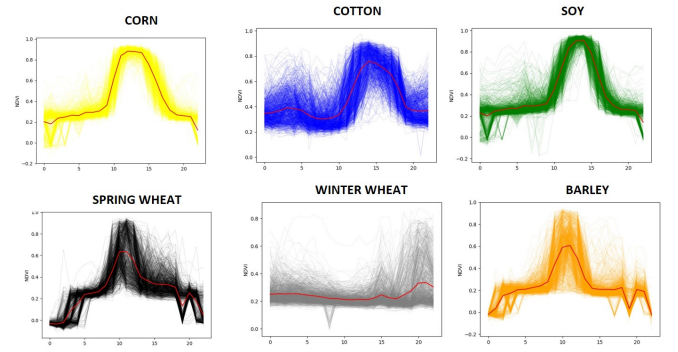


Figure 2: Sample NDVI time series for the crops in our dataset.

images. This is in contrast to a majority of existing methods which employ either purely spatial high-resolution images or purely temporal images in the crop type classification. We use a spatial stream to extract the relevant spatial information from VHR imagery (NAIP, 1m resolution) which has very low temporal resolution and a temporal stream to extract phenological information from high temporal resolution imagery (MODIS, 250m resolution) with coarse spatial resolution. In order to overcome the challenge of variable growth period lengths of each crop, we consider the entire NDVI data for the entire year as our input, with the assumption that the neural network would be able to extract the most relevant features from the entire growth cycle that are relevant to a given crop type.

2 OUR CONTRIBUTIONS

- The Breizh Crop dataset [36] is a purely temporal dataset containing mean aggregates of each of the 13 spectral bands of SENTINEL-2 for field parcels in Brittany, France. While the dataset is large (580k time series, one per field parcel) with high temporal resolution, it has lower spatial resolution compared to VHR imagery such as NAIP, which is only available in the USA. DeepWeeds [29] is high-resolution imagery containing accurate labels for different types of weeds. The data is collected using robotic ground vehicles with cameras mounted 1m above ground level. While this dataset has high spatial information, it seems to lack the high temporal information we are seeking for our experiments. In addition, this data is localized to some select regions where the experiments were conducted and is not based on satellite imagery. As a result, in order to match the requirements of this paper, we provide a curated dataset that contains high spatial resolution images and high temporal resolution time series extracted from multiple locations across the USA. This dataset is described in detail in Section 3. We are planning to release this comprehensive benchmark dataset to the community once additional crop type image collection is completed. In the spirit of reproducible results, the initial data collection used in our experiments in this paper can be downloaded at [16].
- We developed a multimodal classification solution using a two-stream deep neural network that *jointly exploits* the

very high spatial resolution (VHR) imagery (1m NAIP, 3 spectral bands) and high temporal MODIS imagery (250m NDVI, biweekly). Our comprehensive experimental evaluation showed an improved performance of our multimodal solution over the state of the art methods. Code is made available at [15].

- Our work addresses an important societal problem in the area of food system security. As noted earlier, it is important to monitor agriculture lands and produce accurate crop type maps that aid in food production estimates. We hope that both agriculture and machine learning community would benefit immensely from the provided benchmark dataset and accurate classification scheme.

3 DATASET DESCRIPTION

In this section, we describe how different data were collected. The data used in this paper consists of two parts:

- High spatial resolution: 1m resolution NAIP [32] data for the year 2017. NAIP has three spectral bands (Red, Green, Blue).
- High temporal resolution: 250m spatial resolution, bi-weekly temporal resolution Moderate Resolution Imaging Spectroradiometer (MODIS) imagery. We use the Normalized Difference Vegetation Index (NDVI) information from MODIS.

Over 60,000 data points corresponding to the following crops: corn, cotton, soy, spring wheat, winter wheat and barley were collected from multiple locations at different time stamps across the USA. The USDA Cropland Data Layer (CDL) [5], releases 30m resolution classification maps for 255 categories of data for the USA. We used the CDL maps for 2017 as the reference ground truth. Table 1 shows the 6 classes, as well as the states from where their information was collected from. The detailed monthly information regarding the state of each crop (such as planting, flowering, harvesting) is publicly available on the USDA website. The approximate growth cycles shown in Table 1 were defined by roughly considering the time information from planting to harvesting period. The locations were chosen based on the abundance of the specific crops in the states as well as availability of satellite imagery. For example, Iowa and Illinois are well-known for their abundance of corn and soy yields.

As shown in Figure 3, the first phase of data collection is *extraction*, where we traverse the CDL and extract uniform 8×8 patches, where all pixels within the patch belong to the same class (say, $P_1, P_2, \dots$). Given the large volume of information associated with NAIP imagery, we followed a sequential approach, where once the CDL patches were extracted for a particular state, we sequentially parsed through every available NAIP raster and extracted 240×240 patches in such a way that they matched the geospatial locations of each of the P_i . Given that the spatial resolution of NAIP imagery is 1m and that of CDL is 30m, both 8×8 CDL patch and 240×240 NAIP patch capture the same region. Care was taken to ensure that overlapping patches were not selected. At the same time as collecting the high spatial resolution data, the temporal information for each location corresponding to the center of the 8×8 patches P_i specified above is collected. Given the variable growth periods and growing seasons of the crops used in this experiment, the time range for which the data is collected varies. As a result, we collect

one entire years' worth of temporal information for each location. Given the biweekly temporal resolution of MODIS NDVI imagery, this gives us a time series of 23 points for each location. We made use of the python package `gee_subset` [21] which uses the Google Earth Engine (GEE) to extract the MODIS NDVI time series.

The second phase of data collection is *filtration*. As described earlier in Section 1, crops such as winter wheat and spring wheat are grown during different periods, and have different growth cycles. Table 1 demonstrates the states from which each crop has been extracted, as well as the approximate time period for growth cycle for each of the crops (extracted from biweekly crop progress information provided by USDA). The approximate growth cycle is assumed to be all dates between the planting and the harvesting phases for each crop. Care is taken in the filtration step to ensure that the acquisition dates of the high spatial resolution NAIP imagery remain inside the growth cycles of each crop as specified in Table 1.

Both the spatial and temporal data are then split into train, validation and test sets. Table 1 also shows the number of instances of training, validation and test used for our experiments.

4 OUR METHOD

Formally, assume that the dataset D of size n is defined as $D = (X_i, Y_i) \forall i = 1, \dots, n$ and $X_i = (X_i^s, X_i^t)$ and X^s is the high spatial information (NAIP imagery) and X^t refers to the high temporal information (MODIS NDVI imagery) and Y refers to their corresponding class label. Then, the spatio-temporal fusion model $st_fusion()$, which takes the dataset D as input and predicts the class labels can be defined as

$$st_fusion(X^s, X^t) = G(H(F_1(X^s), F_2(X^t))) \quad (1)$$

where the functions F_1 and F_2 act as feature extractors, encoding the spatial-spectral X^s and temporal X^t inputs into high-level feature vectors. H is an aggregation function that fuses the information from F_1 and F_2 and G is a final classification function (for example, MLP or SVM). In the following sections, we describe each of these functions F_1, F_2, H and G in detail.

4.1 Network Architecture

The multimodal spatio-temporal network consists of two streams, spatial and temporal (represented in equation 1 above as F_1 and F_2 respectively).

4.1.1 Spatial Stream. The function F_1 representing the spatial stream is composed of the convolutional layers of a pretrained CNN classification network such as VGG [39], ResNet [17], DenseNet [19] etc., a global average pooling layer and two fully connected layers with ReLU activation (named *S-FC-1* and *S-FC-2*) respectively with a dropout [40] layer in between.

4.1.2 Temporal Stream. The function F_2 representing the temporal stream can be made up of one of the following networks:

- **t-LSTM:** This is made up of a many-to-one LSTM layer with a dropout which is connected to a fully connected layer (named *T-FC-1*) with ReLU activation. Table 2 shows the structure of this network.

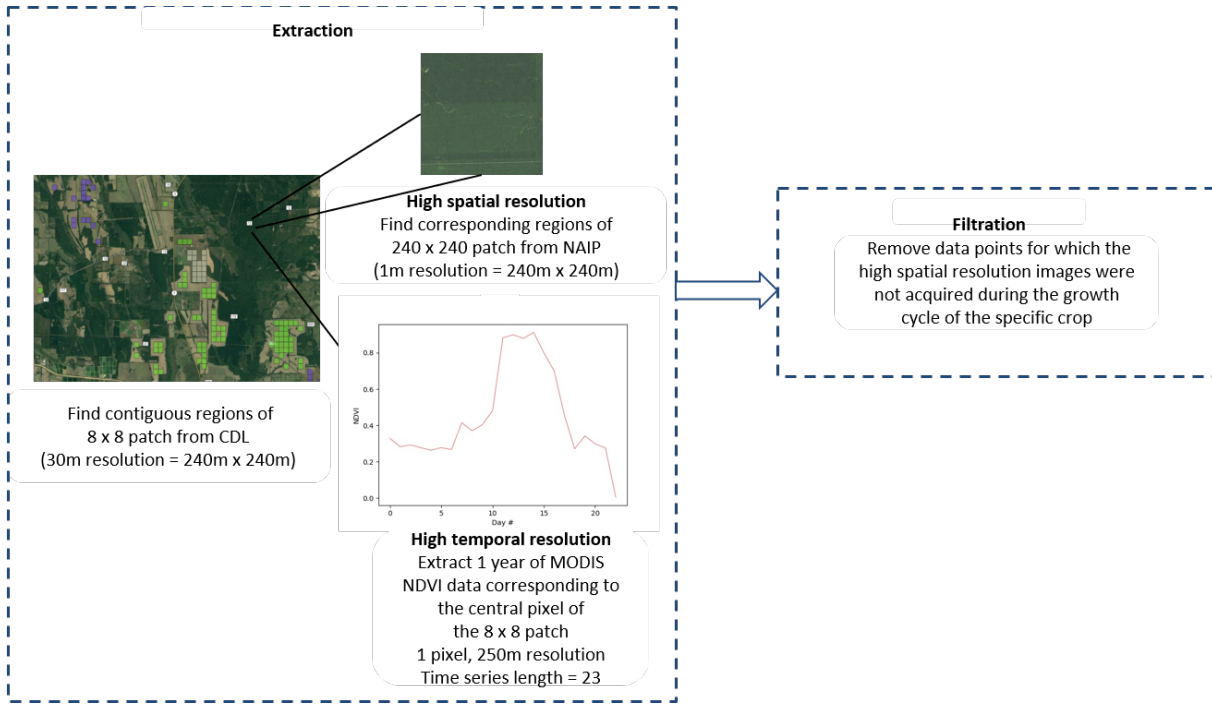


Figure 3: Data collection pipeline

Crop	State(s)	Date range selected	Approximate growth cycle	#Train	#Validation	#Test
Corn	Iowa, Illinois	Jan 2017 - Dec 2017	May 2017 - Oct 2017	6737	2245	2247
Soy	Iowa, Illinois	Jan 2017 - Dec 2017	June 2017 - Oct 2017	6737	2245	2247
Cotton	Georgia, Oklahoma, Alabama	Jan 2017 - Dec 2017	May 2017 - Nov 2017	6737	2245	2247
Spring wheat	North Dakota	Jan 2017 - Dec 2017	Apr 2017 - Aug 2017	6737	2245	2247
Winter wheat	Colorado	Sept 2016 - Aug 2017	Sept 2016 - July 2017	5839	1946	1947
Barley	Montana	Jan 2017 - Dec 2017	Apr 2017 - Aug 2017	3903	1301	1302

Table 1: States from which data was collected

layer name	layer output shape
input layer	batch size x 23 x 1
lstm_1	batch size x 100
T_FC_1	batch size x 32
softmax	batch size x 6

Table 2: Structure of t-LSTM

layer name	layer output shape
input layer	batch size x 23 x 1
bidirectional_lstm_1	batch size x 200
T_FC_1	batch size x 32
softmax	batch size x 6

Table 3: Structure of t-biLSTM

layer name	layer output shape
input layer	batch size x 23 x 1
conv_1 (kernel size = 3)	batch size x 23 x 32
dropout	batch size x 23 x 32
conv_2 (kernel size = 3)	batch size x 23 x 64
average pooling	batch size x 11 x 64
flatten	batch size x 704
dropout	batch size x 704
T_FC_1	batch size x 64
softmax	batch size x 6

Table 4: Structure of t-1D-CNN

- t-biLSTM: Table 3 describes the structure of this network. It is made up of a many-to-one bidirectional LSTM layer with a

dropout, followed by a fully connected layer (named *T-FC-1*) with ReLU activation.

- t-1D-CNN: Table 4 describes the structure of this network. The network is made up of two 1D convolutional layers *conv_1* and *conv_2*, with 32 and 64 feature maps respectively

which are separated by a dropout layer. We apply average pooling along the temporal dimension to the tensor obtained from *conv_2* which is then flattened and passed to a fully connected layer (named *T-FC-1*).

4.1.3 Fusion of Spatial and Temporal Streams. The functions H defined in equation 1 can be represented by combining F_1 and F_2 as described below by using late fusion techniques. Late fusion refers to a strategy where streams of a network are combined close to their output layers. Several methods have been defined to combine two streams, particularly in the area of video action recognition [13, 38]. Figure 4 gives a generic overview of what such a fusion would look like. Here, we describe two ways to fuse the two streams:

- **Concatenation Fusion:** In this approach, H represents the *concatenation* operation that aggregates the outputs of the final fully connected layers of the two streams *S-FC-2* (256 hidden units) and *T-FC-1* (32 hidden units).
- **Average Fusion:** In this approach, H represents the *average* operation that aggregates the the outputs of the final fully connected layers of the two streams *S-FC-2* (32 hidden units) and *T-FC-1* (32 hidden units).

Finally, the outputs of H are processed to obtain the final classification predicted probabilities by using a variety of classifiers such as an MLP or SVM. We describe two ways to represent G as shown below:

- **MLP:** The output of H is passed to an MLP with softmax as the final activation function to obtain the predicted probabilities. The structure of this MLP varies. The MLP can be made up of two fully connected layers (32 hidden units and 6 hidden units respectively) or one fully connected layer (6 hidden units). The first representation will here on be referred to as *ST-C* and a dropout layer is added between the concatenation layer and fully connected layers to avoid overfitting. The second representation will be here on be referred to as *ST-A*.
- **SVM:** In this approach, prior to the concatenation, softmax activation is applied to *S-FC-2* and *T-FC-1*. In other words, H is still concatenation, on a softmax of *S-FC-2* and *T-FC-1*. Then this output of H is passed to an SVM classifier. This network will from here on be referred to as *ST-SVM*.

In Section 4.2, we describe the training procedure for each of the aforementioned network architectures.

4.2 Training

While the high-spatial resolution data is input to the spatial stream, the corresponding high-temporal resolution data is input to the temporal stream. In all our experiments, we use the Adam optimizer [25], together with softmax cross entropy loss. For the spatial network, backbones that were pre-trained on 1000 classes of the ImageNet [11] dataset are used. Given that ImageNet is made up of natural images and the data used in this paper is satellite imagery, we do not freeze any layers in the pre-trained networks and only use the pre-trained weights for initialization. Early stopping [7] is used, and we use the trained weights from the epoch with lowest validation loss for inference.

5 EXPERIMENTS AND RESULTS

We perform three sets of experiments. We first perform an ablation study to prove the effectiveness of both streams in Section 5.1. Then, we describe and compare against several baseline and state of the art methods in 5.2. Finally, we evaluate the fusion strategies described in Section 4.1.3 in 5.3. All the neural network based experiments except for those that use DenseNet201 as the backbone use a batch size of 32, while DenseNet201 uses a batch size of 16. All the neural network based methods are run for a maximum of 32 epochs, subject to early stopping if the validation loss does not improve for more than 8 epochs in sequence. While the neural network based experiments were all implemented using the Keras [10] platform, the baseline machine learning methods were implemented using scikit-learn [30]. For each experiment, test accuracy, macro average F1-score and cohen's kappa score are noted.

5.1 Ablation study on the need for two streams

- **Spatial Stream:** First, the NAIP images collected as described in Section 3 are pre-processed by resizing the 240×240 images to 224×224 to match the input size of the backbone CNNs. In addition, all input images are normalized. Grid search is performed using the hold-out dataset described in Table 1 to identify optimal hyperparameters such as initial learning rate and dropout rate. The optimal learning rate for VGG16 was 0.00001 while it was 0.00005 for ResNet50 and DenseNet. The optimal dropout rate for VGG16 and DenseNet was 0.2 while it was 0.5 for ResNet50. Table 5 reports the overall accuracy and Cohen's Kappa score for the optimal hyperparameters. From the table, it can be seen that VGG16 had the best demonstrated highest test accuracy and kappa score. A similar behavior was also observed in the validation accuracy. Given that we are classifying simple thematic classes such as different types of crops, we chose VGG16 as our backbone network for the spatial stream.
- **Temporal Stream:** First, we pre-process each NDVI time series and remove missing values using nearest neighbor interpolation, followed by standard scaling. Grid search is performed using this pre-processed data to identify the optimal hyperparameters (initial learning rate and dropout rate) for the different types of network architectures described in Section 4.1.2. We observed that both the t-LSTM and t-biLSTM networks achieved a best performance in terms of validation accuracy for an initial learning rate of 0.0005 and no dropout. The t-1D-CNN network achieved its best performance for an initial learning rate of 0.0005 and a dropout rate of 0.1. Table 5 reports the overall accuracy and Cohen's Kappa score for the optimal hyperparameters found. From the table, it can be observed that while the t-LSTM and t-biLSTM both achieved similar performance, the t-1D-CNN achieved a slightly better performance. A similar behavior was observed in the validation accuracies as well. From Tables 2, 3, 4, we can infer the number of trainable parameters in each of these networks. While t-LSTM network has 44,230 trainable parameters, t-biLSTM has 88,230 trainable parameters and t-1D-CNN has 51,846 parameters. Given the not significant increase in the accuracy in comparison to the

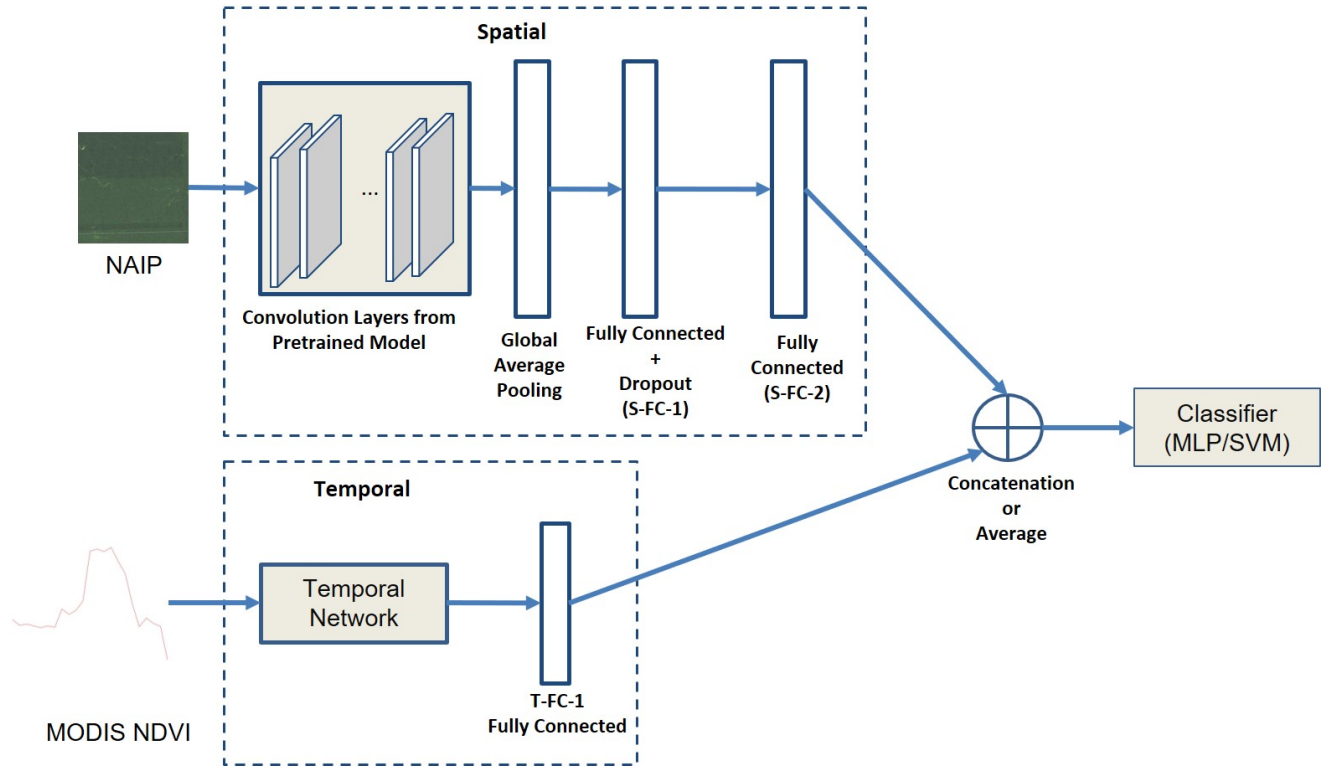


Figure 4: Network structure for fusing the spatial and temporal streams

number of parameters to be trained, we chose t-LSTM as the network to use for the temporal stream.

5.2 Comparison against baseline and state of the art

We re-implemented several purely temporal classification methods described in [42]. All these methods use the MODIS NDVI data described in Section 3. The machine learning baseline methods include Random Forests (RF) and Support Vector Machines (SVM). The LSTM-based and 1D-CNN based networks in [42] (referred to as z-LSTM and z-1D-CNN respectively) serve as our prior art for comparison. Grid search is performed using the validation dataset to identify optimal hyperparameters for each method and the hyperparameters with the highest validation accuracy are chosen. While training, the weights of the epoch with lowest validation loss are saved. It is to be noted that while [42] used additional features such as HANTS [41] and TIMESAT [24] for RF and SVM, they have demonstrated that their 1D-CNN based network outperforms the aforementioned solutions without adding these features. While we use the same parameter grid for RF and SVM for hyperparameter tuning as described in [42], the smaller amount of training data used in this experiment necessitated a difference in parameters such as batch size used for the neural network based methods. While z-LSTM achieved best performance in terms of validation accuracy with an initial learning rate of 0.0005 and a dropout rate of 0.1, z-1D-CNN achieved its best performance with an initial learning rate of 0.00005 and a dropout rate of 0.3. Table 6 reports the overall

accuracy and Cohen’s Kappa score for the optimal hyperparameters. It was observed that the baseline methods had similar performance to the state of the art z-1D-CNN. This can be attributed to the simplicity of the classification task and the limited amount of data available in this experiment in comparison to the large dataset used by [42].

5.3 Fusion of spatial and temporal streams

Finally, we evaluate the fusion strategies described in Section 4.1.3. For the ST-C and ST-A approaches, the training procedure is similar to the procedure described for the spatial and temporal streams in Section 5.1. The high spatial resolution imagery is sent as input to the spatial stream, while the high temporal resolution data is treated as input to the temporal stream. For the ST-SVM, as described in 4.1.3, the predicted class probabilities of the best spatial stream and the best temporal stream models are first collected for the train, validation and test data. Grid search is performed for a variety of hyperparameters for a SVM classifier by using the validation dataset. ST-A achieved best performance with an initial learning rate of 0.0001 and a dropout rate of 0.7, while ST-C achieved best performance with an initial learning rate of 0.0005, a dropout rate of 0.2 between *S-FC-1* and *S-FC-2* and a dropout rate of 0.4 between the concatenation and the first dense layer in the MLP. The inputs to the SVM classifier are from the optimal hyperparameters reported to the VGG16 network and t-LSTM in Table 5. The SVM achieved best performance with an RBF kernel and a gamma of 2 and $C = 1000$. Table 7 shows the evaluation metrics for the three fusion

strategies defined in Section 4.1.3. From the table, we can observe that the best performance was achieved using the ST-SVM method and that it outperformed all the other fusion strategies.

5.4 Discussion

In this section, we compare the methods/models from Tables 5-7. From the tables, it is clearly evident all the two-stream methods performed much better than all the other methods. Figure 5 shows the confusion matrices of the various methods described in Tables 5-7. From the confusion matrix for the purely spatial VGG16 network and the t-LSTM network discussed above, we can notice a large amount of confusion between the pairs corn and soy, and barley and spring wheat. Confusion also persists in the purely temporal z-1D CNN method described above. In comparison, misclassifications between corn and soy are reduced in both ST-Avg and ST-C methods. However, from the confusion matrix of ST-SVM, one can clearly notice a significant improvement in terms of reduced number of misclassifications. This clearly shows that the spatio-temporal approach outperforms the individual purely spatial or purely temporal approaches. Quantitatively, we can say that in comparison to the baseline SVM method and state of the art z-1D-CNN, the prediction error of the best spatio-temporal solution (ST-SVM) has decreased by over 60%.

6 RELATED WORK

Several works have been published in the past two decades in the areas of crop classification and an exhaustive review of literature in this area is beyond the scope of this paper. Most of the research in the area of crop classification differs in terms of the region of interest, the type of satellite imagery used and the methodology. While low, medium and high resolution multispectral imagery (greater than 1m) is available at high temporal resolution, VHR imagery is not available in high temporal resolution. As a result, most works focus on using low, medium and high resolution imagery. [9] presents two per-pixel based approaches (an ensemble of classifiers approach and a co-training based approach) using data collected at different times of the year using the IRS-P6 satellite using the Advanced Wide Field Sensor (AWiFS) camera. [37] introduces a multitemporal approach which combines data from RapidEye and LANDSAT (subsampling to 5m). It integrates phenological models into the classification process by using stochastic Hidden Markov Models. [42] demonstrates a multitemporal approach to crop classification for classifying summer crops using LANDSAT enhanced vegetation index (EVI) using an LSTM and a 1-D CNN, with the 1-D CNN achieving better accuracy. [26] is a multitemporal approach for crop classification using a combination of 4 LANDSAT and 15 SENTINEL-1A images. They compare the performance of an ensemble of 1-D CNNs and 2-D CNNs and demonstrate that the 2-D CNN based solution performs better. [2] demonstrates an object-based multitemporal approach to crop classification using Time Weighted Dynamic Time Warping (TWDTW). [43] presents an object-based approach for crop classification by stacking two SENTINEL-2A images from two different time periods to get a 26-feature vector, and then using a CNN made up of convolutional, pooling and fully-connected layers along the spectral dimension. [35] demonstrates that usage of LSTM networks to extract temporal

properties outperforms traditional classifiers such as SVM. [23] uses a 3D CNN that uses the spectral, temporal and spatial information to perform crop classification. One of the challenges with low and medium resolution imagery is that it is difficult to build accurate crop boundaries with them. In contrast, VHR imagery can assist in building very accurate boundaries. However, VHR imagery has a very low temporal resolution. In contrast to the aforementioned works, our work uses VHR imagery and overcomes its limitation of low temporal resolution by devising a method to combine its high spatial information with another data source which has high temporal resolution. In addition, a majority of the works (with some exceptions) described above are purely temporal in nature, primarily because of the lower spatial resolution of the data under consideration. Also, a direct comparison with [23] is not possible, since structure of the data used in our paper is not suitable to the 3D CNN architecture.

Most closely related to our work, [3] also demonstrated a two-stream neural network architecture with a spatial stream that utilizes VHR imagery (SPOT 5/6, 2m resolution) and a temporal stream that utilizes high temporal resolution imagery (SENTINEL-2, 10m resolution). They use a loss function based on [18] that is a combination of the losses from the spatial, temporal and the concatenated streams. In contrast, our work performs a much more detailed study in terms of the types of networks that can be considered for the spatial or temporal streams, and also performs a more detailed comparison with several baselines. In addition, we also evaluate several fusion strategies and observe that a non-linear combination of the softmax probabilities of the spatial and temporal streams using an SVM classifier gave the best performance.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we have successfully demonstrated the efficacy of a multimodal approach for crop classification. We performed several ablation studies to identify the limitations in using a spatial or a temporal approach alone. We demonstrated several ways of fusing information from the spatial and temporal streams. We compared our approach against the state of the art and demonstrated that the ST-SVM approach reduces the error rate of the state of the art by nearly 60%. While we focus on crop classification in this paper, the central idea of combining high spatial information with high temporal information can also work in other general scenarios. While this work considers the entire growth cycle of a crop, in the future, we would like to extend this work by building methods that are capable of identifying the types of crops early in the crop growth cycle. Research can also be conducted in exploring the use of attention mechanism to identify which part of the growth cycle of a crop is considered most relevant for crop classification. Another promising direction lies in combining information from the spatial and temporal streams right from the early stages in the two-stream network instead of late fusion as performed in this paper. While this paper focuses on using large image patches to match the spatial resolution of the MODIS imagery, combining VHR and HR resolution imagery such as the work proposed by [34] is also worth considering. Our work can be extended in the future to perform yield prediction by adding additional data sources such as climate, weather and soil properties. Finally, while our method

Category/Stream Type	Pretrained Network	Accuracy	Kappa	Macro Avg. F1
Spatial Stream using NAIP	VGG16	95.55	94.62	95.44
	ResNet50	94.92	93.87	94.79
	DenseNet201	95.37	94.41	95.19
Temporal Stream using NDVI	t-LSTM	94.61	93.49	94.82
	t-Bi-LSTM	94.63	93.52	94.80
	t-1D-CNN	95.65	94.75	95.89

Table 5: Performance metrics for ablation study on the test data

Category	Method	Network	Accuracy	Kappa	Macro Avg. F1
purely temporal state of the art on NDVI	z-LSTM [42]	LSTM	93.40	92.03	93.60
	z-1D-CNN [42]	CNN	95.40	94.45	95.60
purely temporal baseline on NDVI	SVM [42]	-	95.65	94.75	95.90
	RF [42]	-	95.26	94.28	95.49

Table 6: Performance metrics for baseline and state of art methods on the test data

Method	Fusion Technique (H)	Final Classifier (G)	Spatial Network (F_1)	Temporal Network (F_2)	Accuracy	Kappa	Macro Avg. F1
ST-C	concatenation	MLP	VGG16	t-LSTM	97.32	96.77	97.42
ST-SVM		SVM	VGG16	t-LSTM	98.41	98.08	98.44
ST-A	Average	MLP	VGG16	t-LSTM	97.16	96.58	97.15

Table 7: Performance metrics for the spatio-temporal fusion strategies on the test data

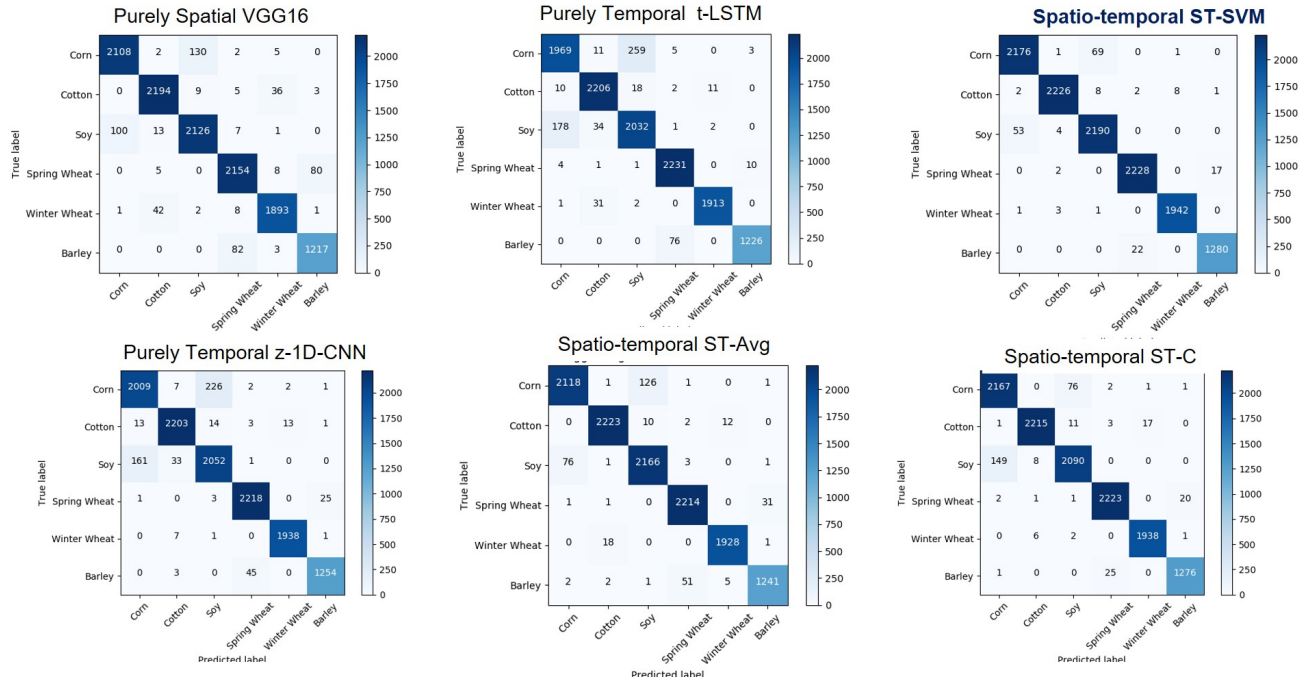


Figure 5: Comparison of confusion matrices of purely spatial VGG16, t-LSTM and ST-SVM methods. Notice the significant improvement in the misclassifications between corn, soy and barley and spring wheat when we go from the purely spatial or purely temporal to spatio-temporal solution

requires the availability of both sources of data, we are working on extending it to situations where one of the sources is not available.

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REFERENCES

- [1] European Space Agency. [n.d.]. GlobCover. http://due.esrin.esa.int/page_globcover.php
- [2] Mariana Belgiu and Ovidiu Csillik. 2018. Sentinel-2 cropland mapping using pixel-based and object-based time-weighted dynamic time warping analysis. *Remote sensing of environment* 204 (2018), 509–523.
- [3] P. Benedetti, D. Ienco, R. Gaetano, K. Ose, R. G. Pensa, and S. Dupuy. 2018. M²Fusion: A Deep Learning Architecture for Multiscale Multimodal Multi-temporal Satellite Data Fusion. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 11, 12 (Dec 2018), 4939–4949. <https://doi.org/10.1109/JSTARS.2018.2876357>
- [4] Jon Atli Benediktsson, Xavier Ceamano Garcia, Bjorn Waske, Jocelyn Chanussot, Johannes R Sveinsson, and Mathieu Fauvel. 2008. Ensemble methods for classification of hyperspectral data. In *IGARSS 2008-2008 IEEE International Geoscience and Remote Sensing Symposium*, Vol. 1. IEEE, 1–62.
- [5] Claire Boryan, Zhengwei Yang, Rick Mueller, and Mike Craig. 2011. Monitoring US agriculture: the US department of agriculture, national agricultural statistics service, cropland data layer program. *Geocarto International* 26, 5 (2011), 341–358.
- [6] Gustavo Camps-Valls, Luis Gómez-Chova, Javier Calpe-Maravilla, Emilio Soria-Olivas, José David Martín-Guerrero, and J Moreno. 2003. Support vector machines for crop classification using hyperspectral data. In *Iberian Conference on Pattern Recognition and Image Analysis*. Springer, 134–141.
- [7] Rich Caruana, Steve Lawrence, and C Lee Giles. 2001. Overfitting in neural nets: Backpropagation, conjugate gradient, and early stopping. In *Advances in neural information processing systems*. 402–408.
- [8] Isabel Luisa Castillejo-González, Francisca López-Granados, Alfonso García-Ferrer, José Manuel Peña-Barragán, Montserrat Jurado-Expósito, Manuel Sánchez de la Orden, and María González-Audicana. 2009. Object-and pixel-based analysis for mapping crops and their agro-environmental associated measures using QuickBird imagery. *Computers and Electronics in Agriculture* 68, 2 (2009), 207–215.
- [9] Varun Chandola and Ranga Raju Vatsavai. 2010. Multi-temporal remote sensing image classification-A multi-view approach.. In *CIDU*. 258–270.
- [10] François Chollet et al. 2015. Keras. <https://keras.io>.
- [11] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *Computer Vision and Pattern Recognition, 2009. CVPR 2009. IEEE Conference on*. Ieee, 248–255.
- [12] O Dubois, JM Faurès, E Felix, A Flammini, J Hoogeveen, L Pluschke, M Puri, and O Unver. 2014. The Water-Energy-Food Nexus: A new approach in support of food security and sustainable agriculture. *Rome, Food and Agriculture Organization of the United Nations* (2014).
- [13] Christoph Feichtenhofer, Axel Pinz, and Andrew Zisserman. 2016. Convolutional two-stream network fusion for video action recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1933–1941.
- [14] Mark A Friedl, Damien Sulla-Menashe, Bin Tan, Annemarie Schneider, Navin Ramankutty, Adam Sibley, and Xiaoman Huang. 2010. MODIS Collection 5 global land cover: Algorithm refinements and characterization of new datasets. *Remote sensing of Environment* 114, 1 (2010), 168–182.
- [15] Krishna Karthik Gadiraju. 2020. Github repository for Multi-Modal Crop Classification. <https://github.com/kkgadiraju/multi-modal-crop-classification>
- [16] Krishna Karthik Gadiraju. 2020. MODIS + NAIP multi-modal dataset for crop classification. <https://bit.ly/2ORb16U>
- [17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.
- [18] Saihui Hou, Xu Liu, and Zilei Wang. 2017. Dualnet: Learn complementary features for image recognition. In *Proceedings of the IEEE International Conference on Computer Vision*. 502–510.
- [19] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. 2017. Densely connected convolutional networks.. In *CVPR*, Vol. 1. 3.
- [20] Alfredo Huete, Chris Justice, and Wim Van Leeuwen. 1999. MODIS vegetation index (MOD13). *Algorithm theoretical basis document* 3, 213 (1999).
- [21] Koen Hufkens. 2017. A Google Earth Engine time series subset script and library. https://github.com/khufkens/gee_subset
- [22] Markus Immitzer, Francesco Vuolo, and Clement Atzberger. 2016. First experience with Sentinel-2 data for crop and tree species classifications in central Europe. *Remote Sensing* 8, 3 (2016), 166.
- [23] Shunping Ji, Chi Zhang, Anjian Xu, Yun Shi, and Yulin Duan. 2018. 3D convolutional neural networks for crop classification with multi-temporal remote sensing images. *Remote Sensing* 10, 1 (2018). <https://doi.org/10.3390/rs10010075>
- [24] Per Jönsson and Lars Eklundh. 2004. TIMESAT—a program for analyzing time-series of satellite sensor data. *Computers & Geosciences* 30, 8 (2004), 833–845.
- [25] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [26] Nataliia Kussul, Mykola Lavreniuk, Sergii Skakun, and Andrii Shelestov. 2017. Deep learning classification of land cover and crop types using remote sensing data. *IEEE Geoscience and Remote Sensing Letters* 14, 5 (2017), 778–782.
- [27] Ajay Mathur and Giles M Foody. 2008. Crop classification by support vector machine with intelligently selected training data for an operational application. *International Journal of Remote Sensing* 29, 8 (2008), 2227–2240.
- [28] Frank Mueller. [n.d.]. ARC A Root Cluster for Research into Scalable Computer Systems. <https://arcb.csc.ncsu.edu/~mueller/cluster/arc/>
- [29] Alex Olsen, Dmitry A. Konovalov, Bronson Philippa, Peter Ridd, Jake C. Wood, Jamie Johns, Wesley Banks, Benjamin Girgenti, Owen Kenny, James Whinney, Brendan Calvert, Mostafa Rahimi Azghadi, and Ronald D. White. 2019. DeepWeeds: A Multiclass Weed Species Image Dataset for Deep Learning. *Scientific Reports* 9, 2058 (14 2 2019). Issue 1. <https://doi.org/10.1038/s41598-018-38343-3>
- [30] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.
- [31] Javier Plaza, Antonio Plaza, and Cristina Barra. 2009. Multi-channel morphological profiles for classification of hyperspectral images using support vector machines. *Sensors* 9, 1 (2009), 196–218.
- [32] National Agriculture Imagery Program. [n.d.]. NAIP National Agriculture Imagery Program. <https://www.fsa.usda.gov/programs-and-services/aerial-photography/imagery-programs/naip-imagery/>
- [33] Global Land Cover 2000 Project. [n.d.]. Global Land Cover 2000 Project. <https://forobs.jrc.ec.europa.eu/products/glc2000/glc2000.php>
- [34] Tim GJ Rudner, Marc Rußwurm, Jakob Fil, Ramona Pelich, Benjamin Bischke, Veronika Kopačková, and Piotr Biliński. 2019. Multi3Net: segmenting flooded buildings via fusion of multiresolution, multisensor, and multitemporal satellite imagery. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 702–709.
- [35] Marc Rußwurm and Marco Korner. 2017. Temporal vegetation modelling using long short-term memory networks for crop identification from medium-resolution multi-spectral satellite images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 11–19.
- [36] Marc Rußwurm, Sébastien Lefèvre, and Marco Körner. 2019. BreizhCrops: A Satellite Time Series Dataset for Crop Type Identification. In *International Conference on Machine Learning (ICML) (Time Series Workshop)*. arXiv:cs.LG, cs.CV, stat.ML/1905.11893
- [37] Sofia Siachalou, Giorgos Mallinis, and Maria Tsakiri-Strati. 2015. A hidden Markov models approach for crop classification: Linking crop phenology to time series of multi-sensor remote sensing data. *Remote Sensing* 7, 4 (2015), 3633–3650.
- [38] Karen Simonyan and Andrew Zisserman. 2014. Two-stream convolutional networks for action recognition in videos. In *Advances in neural information processing systems*. 568–576.
- [39] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014).
- [40] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research* 15, 1 (2014), 1929–1958.
- [41] W Verhoef. 1996. Application of harmonic analysis of NDVI time series (HANTS). *Fourier analysis of temporal NDVI in the Southern African and American continents* 108 (1996), 19–24.
- [42] Liheng Zhong, Lina Hu, and Hang Zhou. 2019. Deep learning based multi-temporal crop classification. *Remote Sensing of Environment* 221, December 2018 (2019), 430–443. <https://doi.org/10.1016/j.rse.2018.11.032>
- [43] Zhuang Zhou, Shengyang Li, and Yuyang Shao. 2018. Object-oriented crops classification for remote sensing images based on convolutional neural network. (2018), 78. <https://doi.org/10.1117/12.2317448>